

Lecture 2

- for most of this course, we'll focus on people who care about two goods, 1 and 2
- so a typical option is a bundle (x_1, x_2) specifying the quantity (x_1) of good 1 and the quantity (x_2) of good 2
- and a utility function specifies the utility (a number) $u(x_1, x_2)$ from every possible bundle (x_1, x_2)
- first part of today:
 - how to graphically represent preferences
 - some examples of preferences
 - utility functions that represent preferences
- second part of today: choice functions and rationalizability

Indifference Curves

Indifference curves are a way to describe preferences over bundles of 2 goods *graphically*:

- an *indifference curve* traces out all points that give a particular “happiness” (utility) level, i.e. among which you’re *indifferent*
- in other words, if you see two points A, B on the same indifference curve, you can infer $A \sim B$
- note: directly graphing the utility function would require a 3D graph, so indifference curves are a more convenient 2D depiction of preferences

Indifference Curve Properties

- rational preferences $\Rightarrow$ indifference curves cannot cross
 - we'll draw this in class: if 2 curves did cross, take points A on the 1st, C on the 2nd, and B at the intersection. Since A and B are on the same curve, $A \sim B$. Similarly, $B \sim C$. Transitivity then yields $A \sim C$, a contradiction since A and C are on different curves.
- if preferences are *monotonic*, indifference curves cannot be thick

Definition

Preferences are monotonic if getting more of some good, without losing anything, makes you strictly better off.

- indifference curves for two “good” goods are downward sloping
 - if you move down (lose some good 2), must move right (gain some good 1) to remain indifferent

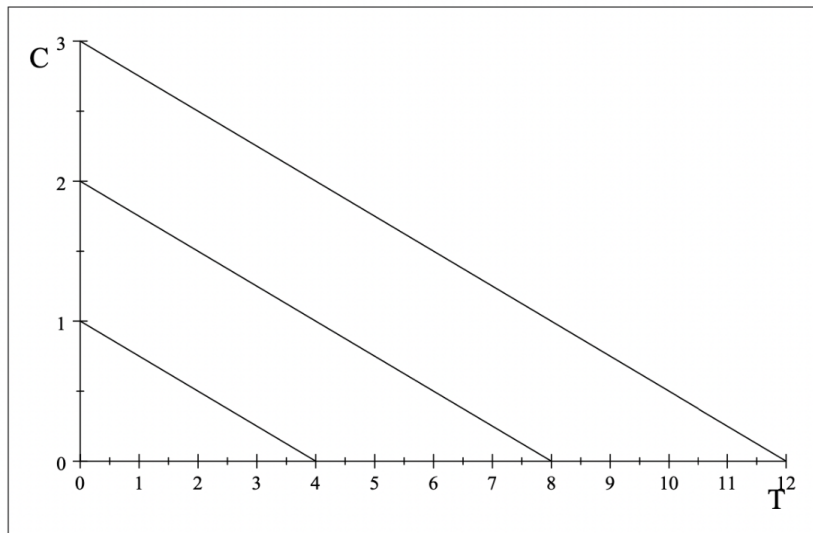
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Special Types of Preferences

Perfect Substitutes

- perfect substitutes are 2 goods where the individual is willing to substitute one for another at a *constant rate*
- e.g., an individual who cares only about caffeine might say "I will trade one coffee for 4 teas, at any consumption level"
- an individual who cares only about having something to write with may be always willing to trade a pen for a pencil
- this constant rate of exchange implies straight-line indifference curves
- in the coffee-tea example, he tells us he's willing to drop one coffee for 4 teas, so with coffee on the vertical axis, indifference curves are straight lines that drop unit for every 4 units across
 - next slide

Perfect Substitute Indifference Curves



Perfect Substitutes Cont'd

- what if the individual will always substitute one pen for one pencil?
- What do the indifference curves look like? Draw a point, and indicate which bundles are better, and which are worse
- **utility representation for pens/pencils?**
 - one: $u(x_1, x_2) = x_1 + x_2$. This says your utility from any bundle of (pens, pencils) is the total # things you have to write with
 - but you could also use say $u(x_1, x_2) = 2x_1 + 2x_2$
 - either way, an indifference curve showing all points with utility $\bar{u}$ is given by $u(x_1, x_2) = \bar{u}$; and if you graph the pairs (x_1, x_2) satisfying this equation, you'll get a straight line with slope -1

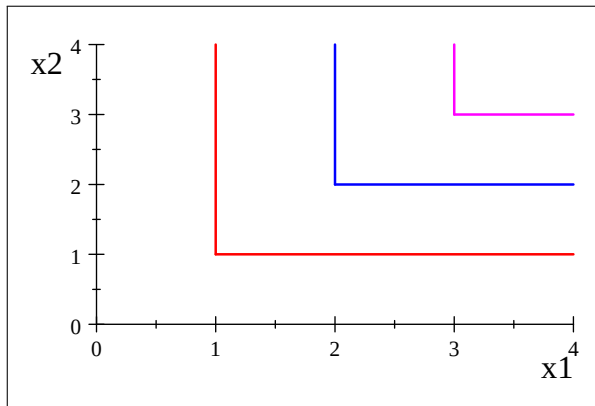
Perfect Complements

- perfect complements are goods that the individual wants to consume in fixed proportions
- "I want one left shoe with every one right shoe", or "I want 2 teaspoons of sugar with every one coffee"
- how do we draw these indifference curves?

Perfect Complements Cont'd

- to draw an indifference curve, pick a point, and then show all points that are exactly as good
- e.g. if we pick $(1,1)$ = (one left shoe, one right shoe), increasing ONLY the # right shoes, or ONLY the # left shoes, leaves him exactly as well off, as it doesn't change the # pairs he has:
- so, all pairs $(1, x)$ or $(x, 1)$ with $x > 1$ are on the same indifference curve
- indifference curve through $(2,2)$ consists of all points $(x, 2)$ or $(2, x)$ with $x \geq 2$: all combinations of (left, right) shoes with 2 right shoes and at least 2 left, or 2 left and at least 2 right
- this suggests a utility function: $u(x_1, x_2) = \min\{x_1, x_2\}$

Perfect Complement Indifference Curves



Marginal Rate of Substitution (MRS)

- the *marginal rate of substitution* (of good 2 for good 1) is the (negative) slope of the indifference curve
- usually, it depends on what point you're looking at:

$$MRS \text{ at } (x_1, x_2) = - \left. \frac{\Delta x_2}{\Delta x_1} \right|_{\text{indifference curve}}$$

- interpretation: it tells you how many units of good 2 you're willing to give up $(-\Delta x_2)$ to get Δx_1 more units of good 1

Marginal Utility and MRS

- with perfect substitutes, it is easy to calculate the slope of the straight-line indifference curves
- but what if they are nonlinear?
 - the slope depends on exactly what point you're looking at
 - for the most accurate measure, use very small changes, i.e. a derivative

Marginal Utility and MRS

- the formula:

$$MRS = - \frac{dx_2}{dx_1} \Big|_{\text{indifference}} = \frac{MU_1}{MU_2}$$

where MU_i indicates “marginal utility”, and is the partial derivative of utility with respect to x_i ($i = 1, 2$)

- proof: along an indifference curve, $u(x_1, x_2) = \text{constant}$, so

$$MU_1 dx_1 + MU_2 dx_2 = 0$$

Marginal Utility Defined

Definition

Marginal utility of good 1 (or 2) is the change in your utility resulting from a small change in the amount of good 1 (or 2).

- observe: a “good” good has a positive marginal utility (getting more of it boosts your utility)
- observe:
 - a steep indifference curve (large MRS) means you’d give up a lot of good 2 for a bit more good 1
 - from our mathematical formula, this happens when MU_1 is large compared to MU_2
 - this makes sense: you’d give up a lot of good 2 if good 1 is much more valuable to you!

MRS: Comprehension Question

- Audrey likes (and has) both cable and Netflix but almost exclusively watches Netflix. Her roommate Adam likes both but watches almost exclusively cable
- sketch their indifference curves over hours of Netflix (vertical axis) and hours of cable (horizontal axis)
- whose are steeper?

Marginal Utility and MRS Example

Assume $u(x_1, x_2) = \log x_1 + x_1^2 x_2 + x_2$. Calculate MU_1 and MU_2 and the MRS. How many units of good 2 will this consumer trade for a unit of good 1 at the point $(1, 2)$?

- if $u(x_1, x_2) = \log x_1 + x_1^2 x_2 + x_2$,

$$MU_1 = \frac{1}{x_1} + 2x_1 x_2$$

$$MU_2 = x_1^2 + 1$$

Marginal Utility and MRS Example

- so,

$$MRS = \frac{MU_1}{MU_2} = \frac{\frac{1}{x_1} + 2x_1x_2}{x_1^2 + 1}$$

- at the point $(1, 2)$ this is $\frac{\frac{1}{1} + 2 \cdot 1 \cdot 2}{1^2 + 1} = \frac{5}{2} = 2.5$, so he's willing to trade 2.5 units of good 2 to get an extra unit of good 1

Preferences and Choices (Motivation)

- if we *know* a consumer's preferences, and we also know which choices are feasible, then we will assume that consumers choose the best feasible option
- if we do not know a consumer's preferences, but we observe his *choices*, we will say that whenever he chooses an alternative, he is revealing that he prefers it to the unchosen feasible alternatives
 - this teases out a *revealed preference* relation
- main goal for rest of today: determine *when* a consumer's choices are *as if* they are choosing optimally with respect to some well-behaved preference relation (a “rational consumer”)

Choice

Definition

A choice rule on a finite set X is a correspondence C that maps each non-empty subset $A \subseteq X$ onto a **non-empty** subset $C(A) \subseteq A$.

- interpretation: $C(A)$ is the set of *all* options that the decision-maker would be willing to choose from A
- typically, A is a set of feasible alternatives (for most of this course, it will be the set of *affordable* consumption bundles)

Rationalizable Choice Rules

- say that a choice rule is *rationalizable* if it could result from a rational consumer who always chooses the best (according to his preferences) feasible option

When are Choices Rationalizable?

- we called a choice rule C *rationalizable* if there exists a (complete and transitive) preference relation $\succsim$ such that *actual* choices agree with the ones that are best, according to $\succsim$
- but, what conditions on choices guarantee (or rule out) existence of such a rationalizing preference?
- we'll present 2 necessary conditions: IIA = Sen's axiom α , and Sen's (consistency) axiom β
- then we'll prove that together, they are equivalent to WARP = Weak Axiom of Revealed Preference, a necessary + sufficient condition for choices to be rationalizable

Independence of Irrelevant Alternatives

(Aka Sen's axiom α , or IIA):

Axiom IIA: If $x \in B \subseteq A$ and $x \in C(A)$, then $x \in C(B)$.

- In words, this says that if you choose x from some “big” set A , then you should also choose it from any subset containing it
- That is, the other options in $A \setminus B$ should be irrelevant to your feelings about x , and so getting rid of them shouldn't change the fact that you consider x a good choice
- For example, it says:

If Simone Biles is the best gymnast in the world, then she must also be the best gymnast in the USA.

Proof: Rationalizability Implies IIA

- take $x \in B \subseteq A$ with $x \in C(A)$
- if C is rationalizable, then choices from any set are optimal according to some preference $\succsim$
- so $x \in C(A) \Rightarrow x$ is a most-preferred choice in A , according to $\succsim$
- But since $B \subseteq A$, it follows that x is also $\succsim$ -optimal in B
- so $x \in C_{\succsim}(B) = C(B)$

Observation II: IIA Does NOT Imply Rationalizability

- let $X = \{a, b, c\}$, and

$$\begin{aligned}C(\{a, b\}) &= \{a\}, C(\{a, c\}) = \{a, c\}, C(\{b, c\}) = \{c\} \\C(X) &= \{a\}, \text{ and } C(\{x\}) = \{x\} \quad \forall x = a, b, c\end{aligned}$$

- C satisfies IIA: a is chosen from X and also all subsets containing a , c is chosen from $\{a, c\}$ and also from $\{c\}$, and b is chosen only from $\{b\}$
- but if C were rationalizable, then $C(\{a, c\}) = \{a, c\}$ would indicate $a \sim c$; this is at odds with the implication from $C(X) = \{a\}$ that $a \succ c$

Sen's "Consistency" Axiom ("Beta")

- the problem with the example above is that it violates Sen's second axiom, "axiom β ":

Sen's axiom β : If $\{x, y\} \subseteq C(B)$, $B \subseteq A$, and $x \in C(A)$, then $y \in C(A)$.

- For example: "If Simone Biles and Aly Raisman are tied for USA champions, and Simone is a world champion, then Aly Raisman must also be a world champion".

Proof: Rationalizability Implies Sen's Beta

- assume $\{x, y\} \subseteq C(B)$, $B \subseteq A$, $x \in C(A)$, and that choices are rationalizable
- so there is a preference $\succsim$ such that $C = C_{\succsim}$
- so x, y are both $\succsim$ -optimal in B , implying $x \sim y$, while x is $\succsim$ -optimal in A
- but then by transitivity, y must also be $\succsim$ -optimal in A , so $y \in C(A)$

Observation: Beta Doesn't Imply Rationalizability

- reason: Sen's axiom β only has any "bite" if there is some set of alternatives from which the DM is willing to choose more than one alternative.
- Thus, take any choice rule in which $C(A)$ is a singleton for every set A , but which violates IIA
- this satisfies β , but is not rationalizable

Next Class

- we'll learn that choices satisfy both α and β if and only if they are rationalizable
- and learn a more applied statement of the α and β axioms together, WARP
- then move on to actually finding best affordable bundles